

LAB-09 UVM TLM & Scoreboard Detailed Notes

LAB-09: Creating a Self-Checking UVM Environment Using TLM and Scoreboards

Objective:

Transform the router environment from stimulus-generation and observation into a self-checking verification environment using TLM, Analysis Ports, Analysis IMPs, Scoreboards, Packet Comparison and Coverage Collection.

Key Concepts:

- Transaction Level Modeling (TLM)
- Analysis Port
- Analysis IMP
- write() Method
- Scoreboard
- Transaction Broadcasting
- Queues
- Packet Cloning
- Packet Comparison
- Self-Checking Verification

What Problem Does the Scoreboard Solve?

Input Packet -> Store Expected Packet

Output Packet -> Compare Automatically

Result -> PASS / FAIL

TLM:

Instead of communicating using signals, clocks and buses, UVM components communicate using transaction objects such as yapp_packet.

Analysis Port:

Declaration:

```
uvm_analysis_port #(yapp_packet) item_collected_port;
```

Creation:

```
item_collected_port = new("item_collected_port", this);
```

Usage:

```
item_collected_port.write(packet_collected);
```

Analysis Port vs Seq Item Port:

Seq Item Port:

Sequencer -> Driver

Supports Handshake

Analysis Port:

Monitor -> Scoreboard

Supports Broadcast Only

Channel Monitor:

Monitors data, data_vld and suspend.

Collects Header, Payload and Parity.

Creates yapp_packet packet_collected.

Broadcasts packet using item_collected_port.write(packet_collected).

Response Collection:
Measures suspend delays.
Creates channel_resp transaction.
Broadcasts response using resp_collected_port.write(response).

Coverage:
Packet Coverage:
ONE = 1
SMALL = 2-10
MEDIUM = 11-20
LARGE = 20-62
MAX = 63

Parity Coverage:
GOOD_PARITY
BAD_PARITY

Response Coverage:
0 cycles
1 cycle
2-10 cycles
11+ cycles

Router Scoreboard:
Purpose:
Compare Expected Packets against Actual Packets.

Architecture:
YAPP Monitor -> Router Scoreboard <- Channel Monitors

Analysis IMP:
Receiver side of TLM.
1 YAPP Analysis IMP.
3 Channel Analysis IMPs.
Total = 4 Analysis IMPs.

Expected Packet Queues:
queue0
queue1
queue2

Routing:
Addr=0 -> Channel0
Addr=1 -> Channel1
Addr=2 -> Channel2

Packet Cloning:
pkt_clone = pkt.clone();

Reason:
Prevents queue corruption due to handle reuse.

Packet Comparison:
expected.compare(actual);

Returns:

1 = Match

0 = Mismatch

Statistics:

packets_received

packets_matched

packets_wrong

Report Phase:

Packets Received

Matched Packets

Wrong Packets

Remaining Queue Entries

TLM Connections:

YAPP Monitor -> Scoreboard

Channel0 Monitor -> Scoreboard

Channel1 Monitor -> Scoreboard

Channel2 Monitor -> Scoreboard

Packet Flow:

YAPP Sequence

-> YAPP Driver

-> Router DUT

-> Channel Monitor

-> Scoreboard

Major Learnings:

Transaction Level Modeling

Analysis Ports

Analysis IMPs

write() Method

Monitor Broadcasting

Scoreboard Design

Queue-Based Checking

Packet Cloning

Packet Comparison

Coverage Collection

Self-Checking Verification Environment

End-to-End Verification

Final Architecture:

Router DUT

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|-- YAPP UVC

|-- Channel UVCs

|-- HBUS UVC

YAPP Monitor + Channel Monitors

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Router Scoreboard

Queue0 Queue1 Queue2

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PASS / FAIL

